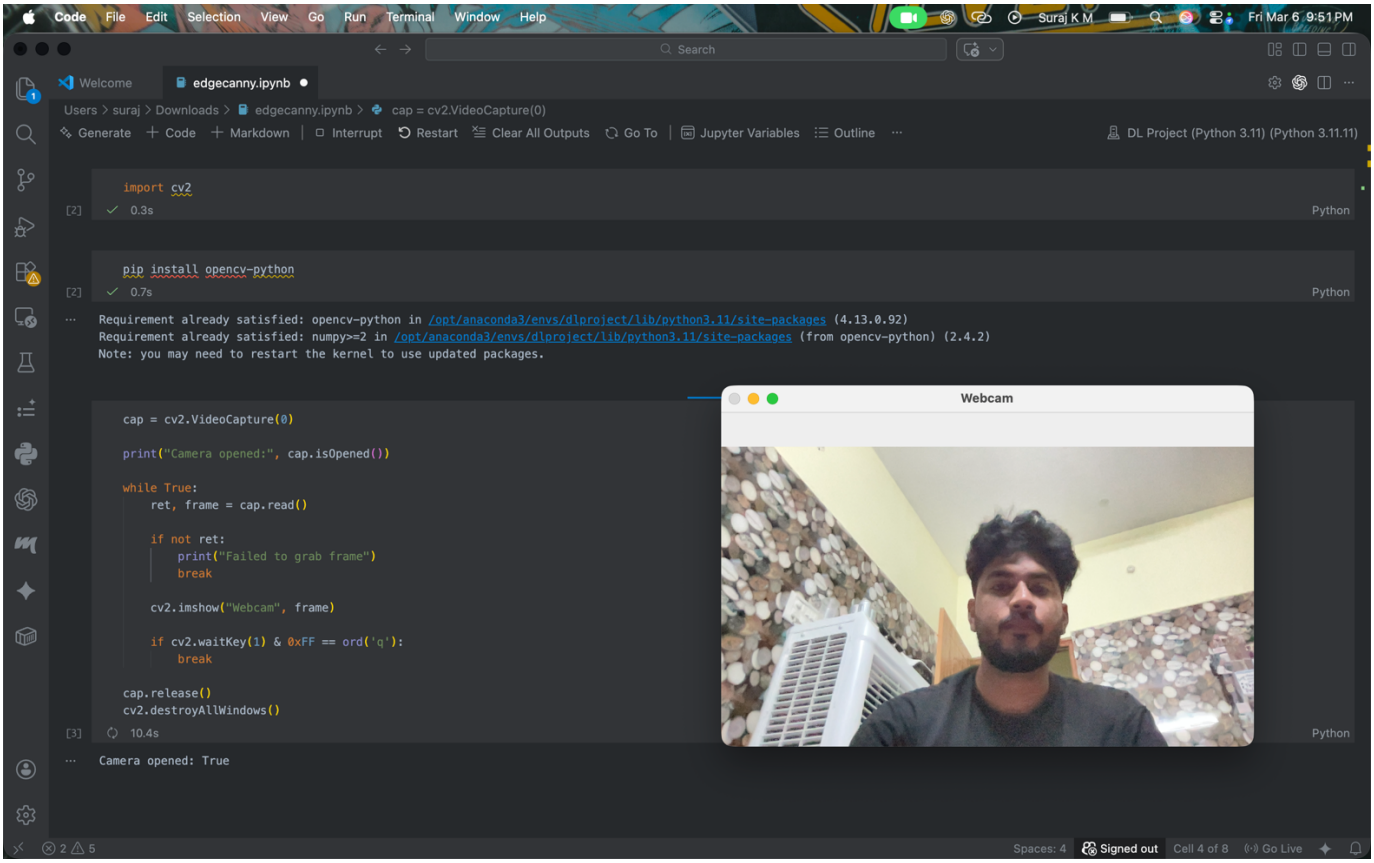
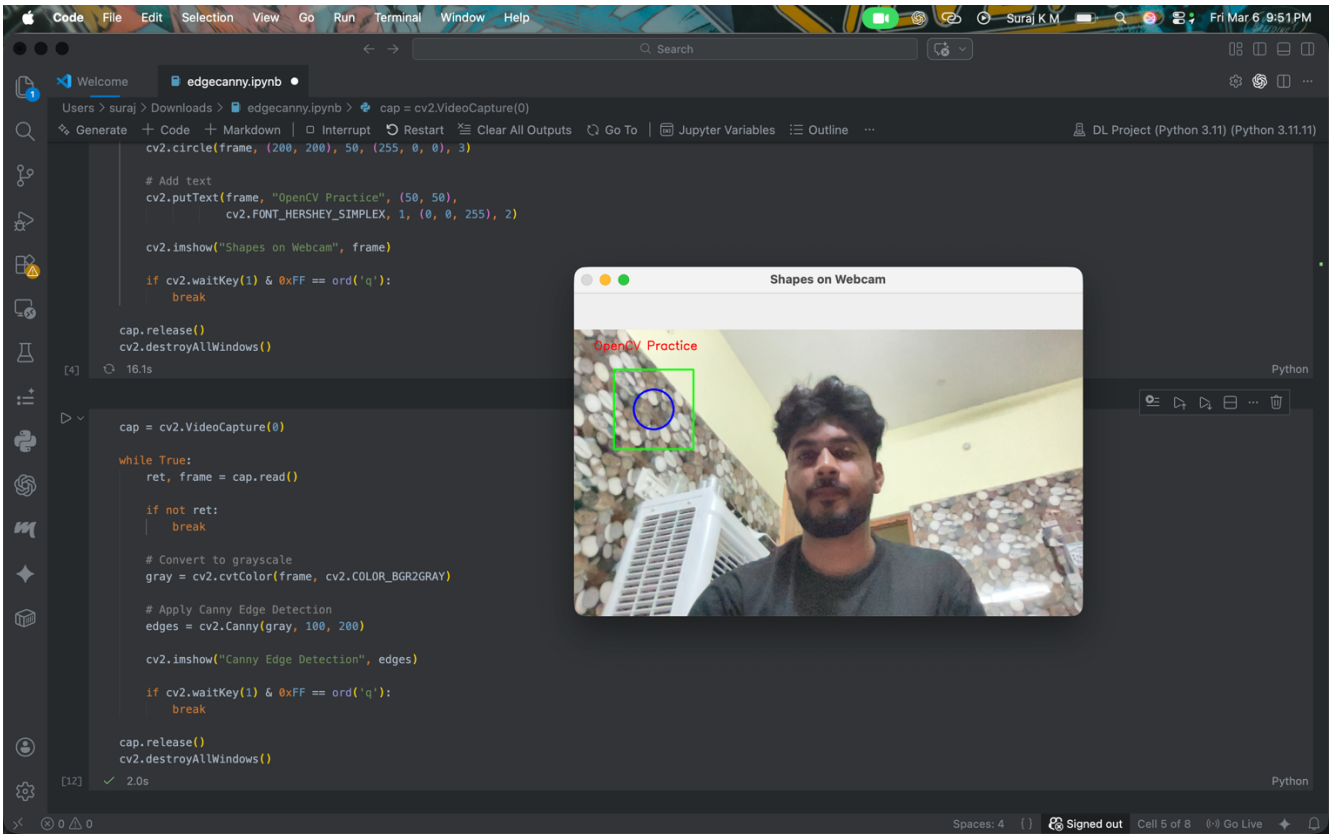
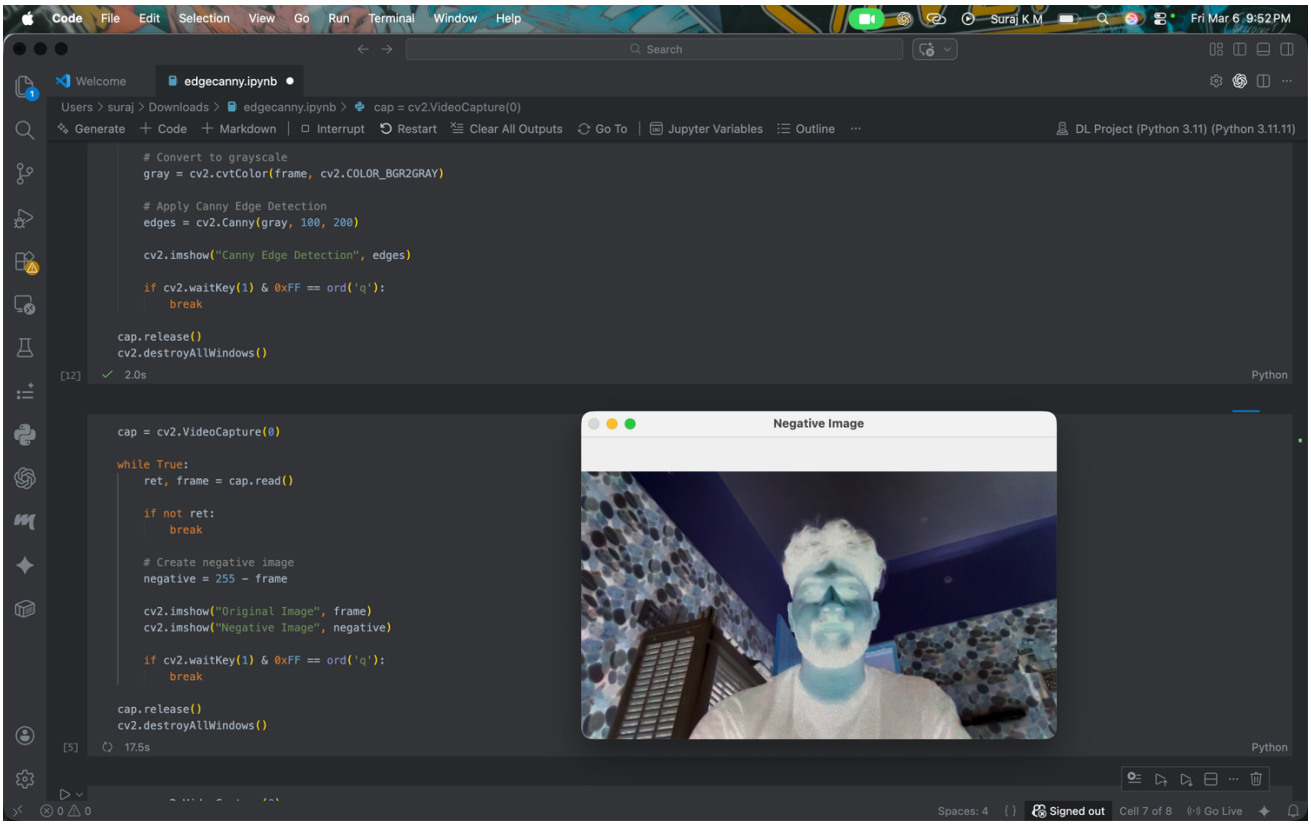
EDGE ASSIGNMENT

SURAJ K M

25MML0047




Code File Edit Selection View Go Run Terminal Window Help
Welcome
edgecanny.ipynb
Users > suraj > Downloads > edgecanny.ipynb > cap = cv2.VideoCapture(0)
Generate + Code + Markdown Interrupt Restart Clear All Outputs Go To Jupyter Variables Outline
DL Project (Python 3.11) (Python 3.11.11)
cv2.circle(frame, (200, 200), 50, (255, 0, 0), 3)
# Add text
cv2.putText(frame, "OpenCV Practice", (50, 50),
cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)
cv2.imshow("Shapes on Webcam", frame)
if cv2.waitKey(1) & 0xFF == ord('q'):
break
cap.release()
cv2.destroyAllWindows()
[4] 16.1s
Python
cap = cv2.VideoCapture(0)
while True:
ret, frame = cap.read()
if not ret:
break
# Convert to grayscale
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
# Apply Canny Edge Detection
edges = cv2.Canny(gray, 100, 200)
cv2.imshow("Canny Edge Detection", edges)
if cv2.waitKey(1) & 0xFF == ord('q'):
break
cap.release()
cv2.destroyAllWindows()
[12] 2.0s
Python
Shapes on Webcam
OpenCV Practice
Spaces: 4 Signed out Cell 5 of 8 Go Live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Welcome
edgecanny.ipynb
Users > suraj > Downloads > edgecanny.ipynb > cap = cv2.VideoCapture(0)
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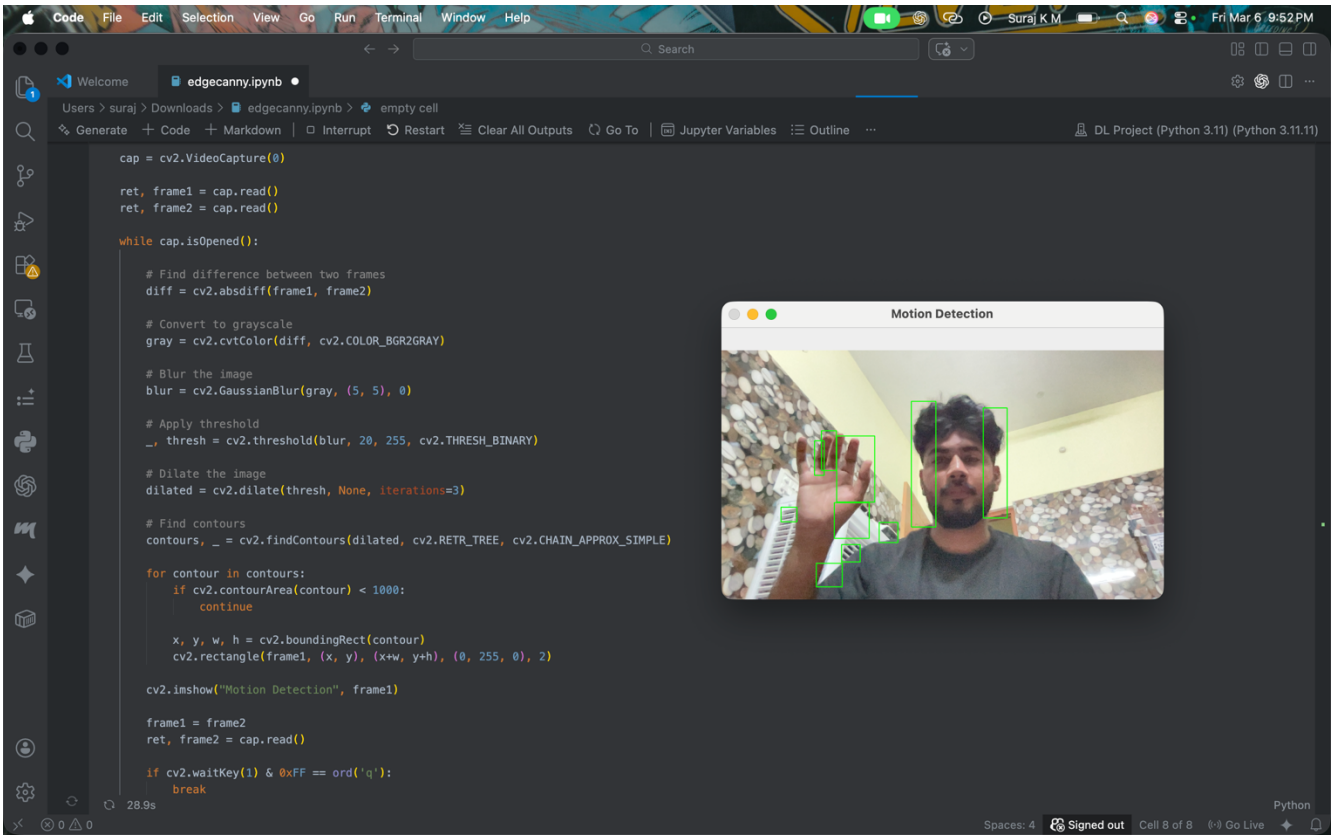
DL Project (Python 3.11) (Python 3.11.11)
# Convert to grayscale
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
# Apply Canny Edge Detection
edges = cv2.Canny(gray, 100, 200)
cv2.imshow("Canny Edge Detection", edges)
if cv2.waitKey(1) & 0xFF == ord('q'):
break
cap.release()
cv2.destroyAllWindows()
[12] 2.0s
Python
cap = cv2.VideoCapture(0)
while True:
ret, frame = cap.read()
if not ret:
break
# Create negative image
negative = 255 - frame
cv2.imshow("Original Image", frame)
cv2.imshow("Negative Image", negative)
if cv2.waitKey(1) & 0xFF == ord('q'):
break
cap.release()
cv2.destroyAllWindows()
[5] 17.5s
Python
Negative Image
Spaces: 4 Signed out Cell 7 of 8 Go Live


Code File Edit Selection View Go Run Terminal Window Help
Suraj K M
Fri Mar 6 9:52 PM
Search
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DL Project (Python 3.11) (Python 3.11.11)
cap = cv2.VideoCapture(0)
ret, frame1 = cap.read()
ret, frame2 = cap.read()
while cap.isOpened():
# Find difference between two frames
diff = cv2.absdiff(frame1, frame2)
# Convert to grayscale
gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)
# Blur the image
blur = cv2.GaussianBlur(gray, (5, 5), 0)
# Apply threshold
_, thresh = cv2.threshold(blur, 20, 255, cv2.THRESH_BINARY)
# Dilate the image
dilated = cv2.dilate(thresh, None, iterations=3)
# Find contours
contours, _ = cv2.findContours(dilated, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)
for contour in contours:
if cv2.contourArea(contour) < 1000:
continue
x, y, w, h = cv2.boundingRect(contour)
cv2.rectangle(frame1, (x, y), (x+w, y+h), (0, 255, 0), 2)
cv2.imshow("Motion Detection", frame1)
frame1 = frame2
ret, frame2 = cap.read()
if cv2.waitKey(1) & 0xFF == ord('q'):
break
28.9s
Motion Detection
Python
Spaces: 4 Signed out Cell 8 of 8 Go Live